ROHINI SHARMA



Associate Professor
Department of Computer Science and Applications
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Personal Details

Father's Name	Roshan Lal Sharma
Date of Birth	20-08-1981
Marital Status	Married
Address	E-16, Sector 14, Panjab University Campus, Chandigarh.

Educational Qualifications

- Ph.D in 2019
- M.Tech.(CSE) - CGPA 9.5 in 2011
- UGC-JRF qualified in June, 2008
- MCA - 72% (approx.) in 2005
- BCA - 78% (approx.) in 2002
- 10+2 (Non-Medical) in 1999 (Awarded State Merit Scholarship)

Experience

September, 2024 till date	Associate Professor (Academic Level 13A), DCSA, PU, Chd.
September, 2021 to September, 2024	Assistant Professor (Selection Grade/Academic Level 12), DCSA, PU, Chd.
September, 2016 to September, 2021	Assistant Professor (Senior Scale/Academic Level 11), DCSA, PU, Chd.
September, 2011 to September, 2016	Assistant Professor, Stage-I, DCSA, PU, Chd.
July, 2009-September, 2011	Programmer, GNDU, Amritsar.
July, 2007 – Nov., 2007	Lecturer (Ad-hoc), GNDU, Amritsar
Sept., 2006 – March, 2007	Lecturer (Ad-hoc), GNDU, Amritsar

Administrative Responsibilities

July 7, 2021 – July 6, 2024	Chairperson
August 6, 2020 – Jan. 17, 2021	Chairperson (Officiating)
July 03, 2020 – July 31, 2020	Chairperson (Officiating)
July 7, 2021 – July 6, 2024	Co-ordinator, ADCA, Panjab University

Ph.D Thesis

Title: Flow based Anomaly Detection and Behavioral Analysis in Computer Networks
Institution: Panjab University, Chandigarh
Supervisors: Prof. R.K. Singla, Dr. Ajay Guleria

Research Guidance (Ph.D)

- Ms. Rashmi Bhatia, Enrolled in 2019
- Ms. Manpreet Kaur, Enrolled in 2019
- Ms. Rajbinder, Enrolled in 2019
- Ms. Vikramjeet, Enrolled in 2023

Research Interests

- Network Traffic data analysis
- Machine Learning
- Natural Language Processing

Research Papers Published

- Bhatia, R., and Sharma, R. Layered feature optimization framework based on Filtering, Embedding and data Balancing (L-FEB) for efficient DDoS attack detection. *Expert Systems with Applications*, *277*, 127230 (2025). https://doi.org/10.1016/j.eswa.2025.127230 **(IF – 7.5)**
- Kaur, R., Dhaliwal, M. K. and Sharma, R. Improving Detection of Falls and Activities Using Machine Learning Model. In Proceedings of International Joint Conference on Computing Sciences (ICCS-2023) (2024).
- Kaur, R., Sharma, R. Wearable sensors and datasets for evaluating systems predicting falls and activities of daily living: recent advances and methodology. Multimedia Tools Applications 83, 73671–73701 (2024). https://doi.org/10.1007/s11042-024-19504-1 **(IF - 3)**
- Dhaliwal, M. K., Sharma, R. Improving Accessibility and Independence for Blind/Visually Impaired Persons based on Speech Synthesis Technology. International Journal of Computer Applications 186(28), 12-20 (2024). https://doi.ort/10.5120/ijca2024923768
- Bhatia, R. and Sharma, R. Multiclassification of DDoS attacks using machine and deep learning techniques. International Journal of Security and Networks, 19(2), 63-76 (2024). https://doi.org/10.1504/IJSN.2024.140268
- Dhaliwal MK, Sharma R, Kaur R. PIF dataset: a comprehensive dataset of physiological and inertial features for recognition of human activities. Multimedia Tools and Applications 2, 1-9 (2024). https://doi.org/10.1007/s11042-024-19285-7 **(IF - 3)**
- Kaur R, Sharma R, Dhaliwal MK. Physiological and Inertial Features based Dataset for Falls and Activities: PIF v2. Procedia Computer Science 1 (235), 1268-77 2024. https://doi.org/10.1016/j.procs.2024.04.120
- Kaur, R., Sharma, R., Dhaliwal, M.K. Evaluating Performance of SMOTE and ADASYN to Classify Falls and Activities of Daily Living. In Proceedings of the 12th International Conference on Soft Computing for Problem Solving. SocProS 2023. Lecture Notes in Networks and Systems, vol 995. Springer, Singapore (2024). https://doi.org/10.1007/978-981-97-3292-0_22
- Dhaliwal, M.K., Sharma, R., Bindra, N. Analyzing Wearable Data for Diagnosing COVID-19 Using Machine Learning Model. Lecture Notes in Electrical Engineering, vol 946. Springer, Singapore. https://doi.org/10.1007/978-981-19-5868-7_22
- Dhaliwal, M.K., Sharma, R., Bindra, N. Role of Internet of Things (IoT) in Preventing and Controlling Disease Outbreak: A Snapshot of Existing Scenario. In Proceedings of the International Conference on Intelligent Computing, Communication and Information Security. ICICCIS 2022. Algorithms for Intelligent Systems. Springer, Singapore (2023). https://doi.org/10.1007/978-981-99-1373-2_28

- Sharma R, Guleria A, Singla R. K. Flow-based profile generation and network traffic detection for DNS anomalies using optimised entropy-based features selection and modified Holt Winter's method. International Journal of Security and Networks 16 (4), 244-57 (2021). https://doi.org/10.1504/IJSN.2021.119380
- Bhatia, R., Sharma, R., Guleria, A. Anomaly Detection Systems Using IP Flows: A Review. In Advances in Clean Energy Technologies. Springer Proceedings in Energy. Springer, Singapore (2021). https://doi.org/10.1007/978-981-16-0235-1_80
- Sharma, R., Guleria, A., Singla, R.K. Characterizing Network Flows for Detecting DNS, NTP, and SNMP Anomalies. In Advances in Intelligent Systems and Computing, vol 673. Springer, Singapore (2018). https://doi.org/10.1007/978-981-10-7245-1_33
- Sharma R, Singla R. K, Guleria A. A new labeled flow-based DNS dataset for anomaly detection: PUF dataset. Procedia computer science 1(132), 1458-66 (2018). https://doi.org/10.1016/j.procs.2018.05.079
- Sharma R, Guleria A, Singla R. K. An overview of flow-based anomaly detection. International Journal of Communication Networks and Distributed Systems 21(2):220-40 (2018). https://doi.org/10.1504/IJCNDS.2018.094221 **(IF - 1)**
- Sharma, R. Internet of Things: An approach for advancement in educational institution. 1st India International Conference on Information Processing (IICIP), Delhi, India, 1-4 (2016) doi: 10.1109/IICIP.2016.7975351.
- Sharma, R. Performance Evaluation of Frequent Pattern Mining Techniques. Panjab University Research Journal (2016).
- Sharma, R. Frequent Pattern Mining Based System for Efficient Anomaly Detection. Panjab University Research Journal (2016).

Research Papers Presented

- “Developing IMU and Physiological Signals based Systems for Activities and Fall Detection” in International Conference on Computing Advances in AI and ML organized by Department of Computer Science and Applications, Panjab University, Chandigarh from 16-17, December, 2024.
- “Layered Feature Optimization framework for efficient DDoS attack detection System” in CHASCON-2024 National Conference on Indigenous Technologies for Viksit Bharat’ organized by Panjab University from 6-8 November, 2024.
- “Analysis of UP-Fall Dataset using SMV and Cos Angles” in International Conference on Artificial Intelligence and Machine Learning organized by the Department of Computer Science and Applications, Panjab University, Chandigarh from November 28th - 29th, 2023.
- “Statistical And Machine Learning Based Analysis on GSR Dataset” in International Conference on Artificial Intelligence and Machine Learning organized by the Department of Computer Science and Applications, Panjab University, Chandigarh from November 28th - 29th, 2023.
- “Review of IoT Based Systems in Epidemics” in 14th Chandigarh Science Congress organized by Panjab University, Chandigarh in association with Chandigarh Region Innovation and Knowledge Cluster (CRIKC), 2020.
- "Entropy based Features Selection Technique for Network Anomaly Detection” in Chandigarh Science Congress-2019. (Awarded best oral presentation award)
- “Open Source Digital Repository Softwares – A review” in CHASCON-2012.

- "A survey of emerging trends in Vehicular Ad-hoc Networks" International Conference on Technology and Information Systems (ICTIS) (2010).

Keynote Speaker / Resource Person / Talks delivered

- Workshop on Artificial Intelligence & Machine Learning organized by UIAMS, Panjab University, Chandigarh in April, 2024.
- National webinar on "Cyber Security and its Ethical Considerations" conducted on 11-March-2024 by Gujranwala Guru Nanak Khalsa College, Ludhiana.
- CSDE Skill Enhancement Summer Course on 'Data Analysis and Visualization using Python' organized by Centre for Skill Development and Entrepreneurship, Panjab University, Chandigarh in July, 2023.
- Skill Enhancement Workshop/Course on 'Python Programming' organized by Rashtriya Uchchtar Shiksha Abhiyaan (RUSA) in collaboration with Department of Computer Science and Applications(DCSA), Panjab University, Chandigarh in April, 2023.
- Two days National Seminar on "Recent Advances in Growth and Prospects of Digital India: Revolutionizing Education through Digitalization (RAGPDI-RETD-2023) organized by PCM SD College for Women, Jalandhar in March, 2023
- Skill Enhancement Workshop/Course on 'Introduction to Data Science' organized by Department of Computer Science and Applications(DCSA) under the aegis of Rashtriya Uchchtar Shiksha Abhiyan (RUSA), Panjab University, Chandigarh in February, 2023.
- Webinar on "Supervised Learning in Data Analytics" conducted on 03-Sep-2022 by Gujranwala Guru Nanak Khalsa College, Ludhiana.
- Seven days National Workshop (online) on "Open Source Software Utility to applied Econometrics and Statistics" on 24-August-2022 organized by Department of Economics in collaboration with Department of Computer Science and Applications and Department of Statistics.
- Induction Course for Newly recruited PGT(CS/IT) on 30-June-2022 organized by Navodaya Leadership Institute Bhillowal, Amrtisar held at NLI Sub-Campus, Chandigarh.
- Web-series season 2 on "Evolving Technologies and recent trends in Computer Science" on 23-02-2022 organized by DAV University, Jalandhar, Punjab.
- Webinar on 'Evolving technologies and recent trends in Computer Science' on 25-02-2021 organized by DAV University, Jalandhar, Punjab.
- Faculty Development Program organized jointly by DCSA and CALEM on 'Artificial Intelligence and Machine Learning: Basics and Applications' from 22 – 28 December, 2020.
- Workshop on Excel in Chandigarh University, Gharuan, Mohali in November, 2016
- Natural Language Processing to PGT Computer Science in service teachers in Zonal Institute of Education and Training, Kendriya Vidyalaya Sangathan in 2015.

Seminars / Workshops / Webinars/ Faculty Development Programs/ Events / Activities Convened / Co-ordinated

- Coordinator, One day National Workshop on 'Natural Language Processing' organized by DCSA on 27th March, 2024

- Coordinator, International Conference on Artificial Intelligence and Machine Learning organized by DCSA on 28-29 November, 2023
- Coordinator, 15-days Skill Enhancement Workshop on "Introduction to Data Science' from 1st – 21st February, 2023
- Section Co-ordinator of Mathematical Science Section in Chandigarh Science Congress (CHASCON), September, 2022
- Seven days national workshop (online) on "Open Source Software Utility to applied Econometrics and Statistics" from 19-August-2022 to 25-August-2022 in collaboration with Department of Economics and Department of Statistics.
- One day Symposium of "Computational Sciences" on 25-August-2022 in collaboration with Department of Statistics.
- One day seminar on "Universal Values: Ethics in Education and Education in Ethics" on 22-August-2022 in collaboration with Department of Mathematics and Department of Statistics.
- Tree Plantation drive on 17-August-2022 in collaboration with Department of Mathematics and Department of Statistics.
- Virtual Cycling Event on 14 and 15-August-2022 and felicitation ceremony on 02-September-2022 to celebrate 75 years of India's Independence.
- Convenor, National Seminar on 'Commemoration of World Intellectual Property day 2023' on 26th April, 2023 organized in collaboration with DST-Centre for Policy Research, DPIIT-IPR Chair at Panjab University (PU), Chandigarh and Knowledgentia Consultants, New Delhi
- Coordinator, One Day workshop on "Machine Learning Techniques" organized by DCSA on 04-Feb-2022.
- Bicycle Rally: An Initiative towards Green Computing organized by DCSA in collaboration with NSS of PU and Boys Hostel No. 2 on 12-Dec-2021.
- Webinar on 'Opportunities and Challenges of Machine Learning based solutions' on 27 October, 2021
- Webinar on 'Network Security Analytics' on 13 August, 2021.
- One week interdisciplinary Faculty Development Programme organized jointly with CALEM on 'Artificial Intelligence and Machine Learning: Basics and Applications' from 22 – 28 December, 2020.
- Expert Talk and Section presentations in the Department in Chascon, 2020.
- Webinar on 'Building up Happy and Positive Minds during Pandemic times' on 10 November, 2020
- Online E – poster making competition in December, 2020.
- Online Centralized Counselling for admission to MCA and M.Sc. in DCSA, Regional Centre Hoshiarpur and Regional Centre, Muktsar.
- Plantation Drive in DCSA in August, 2020.
- Webinar on 'National Education Policy 2009/2020: Open Standards for Interactive Computing' on 15 October 2020.
- Faculty Development Programme organized jointly with Human Resource Development Centre, Panjab University on 'Artificial Intelligence in Human Life' from 6 to 12 October, 2020
- Webinar on 'National Education Policy 2020: Introducing Coding at young age" on 25 September 2020
- Worked as Section Secretary in CHASCON-2018 and CHASCON-2019

- Course Coordinator for Pre-Ph.D course-work in the Department of Computer Science and Applications in 2019
- Time-table Co-ordinator in the Department of Computer Science and Applications from June, 2014 to December, 2019.
- Assistant Co-ordinator in One day Workshop on “Syllabus revision for Under Graduate and Post Graduate courses for Computer Science and Information Technology” organized by Department of Computer Science and Applications, PU, Chandigarh in 2015
- Co-ordinated number of departmental activities including Orientation-cum-Induction program for Freshers (2019), IT Fest (2018, 2017, 2015, 2014), Women Day Celebration (2020) etc.

Orientation/ Refresher / Faculty Development Programs attended

- Short Term Course on ‘Advancements in Healthcare Domain using AI and ML Techniques’ organized by NIT, Kurukshetra from 6-10, January, 2025.
- Faculty Development Program organized by ABV-IIITM, Gwalior from 2-8 August, 2024.
- Faculty Development Program organized by SMVD University, Katra, Jammu from 8-13 July, 2024.
- NEP 2020 Orientation and Sensitization Program organized by UGC, Mission Malaviya Teacher Training Centre, NIEPA, New Delhi from 20-29 November, 2023.
- Refresher Course in Information Technology organized by DCSA and HRDC, PU, Chandigarh in February, 2021
- Faculty Development Program on ‘Managing Virtual Classrooms and Open Educational Resources’ in June, 2020
- Short Term Course on "Internet of Things: Application Development using Raspberry Pi and Aurdino Boards" in 2019.
- Refresher Course in Research Methodology (Science) organized by Department of Botany and HRDC, PU, Chandigarh in 2019.
- Short Term Course on Python in NITTTR, Chandigarh in 2018.
- Faculty Development Program on "Big Data and Data Analytics" in UIET, PU Chandigarh in 2016
- Refresher Course in Information Technology organized by Department of Computer Science and Applications and HRDC, PU Chandigarh in 2015
- Orientation Course for Teachers organized by HRDC, PU Chandigarh in 2013

Seminars and Workshops attended

- Online session on Virtual Labs organized by UIET, Panjab University, Chandigarh in April, 2020
- Workshop on "Machine Intelligence and Deep Learning: An interdisciplinary perspective" organized by CSIO in 2019.
- One day seminar on "Role of ICT in Education, Research and Innovation" organized by Department of Computer Science and Applications in 2014
- Two days Symposium on “Curriculum Revision and Quality Education” held in DCSA, PU, Chandigarh in 2013.

- One day workshop on "e-Governance and National e-Governance Plan" organized by Computer Centre, Panjab University, Chandigarh in 2013.
- Seminar on "Awareness on e-Governance and National e-Governance plan organized by Computer Centre, Panjab University, Chandigarh sponsored by Department of Electronics and Information Technology, Ministry of Communication and Information Technology, Government of India in December 2012.
- One day Symposium on "Recent Trends in Computational Techniques" organized by the Department of Computer Science and Applications, Panjab University, Chandigarh in December 2012.
- Training program on "Network Simulation using NS-2" from ATOSS Educational Society, Amritsar in 2011.
- Short term program on "Open Source Technologies" conducted by NITTTR Chandigarh in 2010
- Workshop on "Ethical Hacking and Information Security" conducted by Kyrion Digital Securities at NIT Jalandhar in 2009.

Session Chair in Conference / Seminar

- Session Chair in All India Seminar on "Business Intelligence with Intelligent Computing" organized by The Institute of Engineers (India), Punjab and Chandigarh State Centre & BAM Khalsa College, Garhshankar, Hoshiarpur on 05-03-2022.
- Session Chair in 1st International Conference on Innovations in Clean Energy Technology (ICET-2020) organized by MANIT, Bhopal in 2020.

Membership of Societies / Committees

- Life Member of Computer Society of India (CSI)
- Member, International Association of Engineers, Hong Kong.
- Acted as member of Board of Studies (BoS) of School of Computing, Lovely Professional University, Jaladhar in April, 2023.
- Member, Skill Development Board of Panjab University, Chandigarh (July 2021 to July 2024)
- Member of RAC Committee of DCSA.
- Member of RDC Committee of DCSA (July 2021 to July 2024)
- Member of Board of Control of DCSA, PU, Chandigarh (Jan, 2015- Dec, 2017, Jan, 2020 - till date).
- Member of Academic, Administrative and Technical Committee of DCSA, PU, Chandigarh.
- Member, Placement Cell, DCSA, PU, Chandigarh (Jan-2017 to Dec-2018).
- Member, Anti-ragging Committee, PU, Chandigarh (2015)

Subjects Taught

- Analysis and Design of Algorithms
- Software Engineering
- Numerical and Statistical Methods
- Artificial Intelligence
- Programming in C/C++

- Computer Networks
- Computer Fundamentals
- Data and File Structures